TUTORIAL

Conformational change with NEB: cyclohexane

> Updated for EasyHybrid 3.2.0 (2026-10-26). The figures and numbers come from a run of this tutorial with the 3.2.0 development version.

A scan needs a reaction coordinate. The **nudged elastic band** (NEB) method needs only the two end structures: it builds a chain of intermediate structures (images) between them, joined by springs, and optimizes the whole chain until each image lies on the lowest energy path. The highest image approximates the transition state.

Here NEB follows the ring flip of cyclohexane from the **chair** to the **twist-boat**, a change of all six ring torsions at once that no single coordinate describes well.

The structures are in `examples/tutorials/05_neb_cyclohexane/` of the EasyHybrid folder: `cyclohexane_chair.mol` and `cyclohexane_twistboat.xyz` (the same atom order in both files). The whole tutorial takes about one minute.

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1. Create the system and the two end structures

1. **File → New**, **System Type** *DYFF*, file `cyclohexane_chair.mol`, name `cyclohexane` (Figure 1). **Import System**.
2. **Simulate → Geometry Optimization** of `0_coords` (RMS gradient 0.01): the optimized chair, `0_Geometry_Optimization_0`.
3. **File → Import Data...**: format *xyz – coordinate file*, the file `cyclohexane_twistboat.xyz`, **As a new object** `twistboat` (Figure 2). The coordinates are added to the same system, as a new object.
4. Optimize `twistboat` too: `0_Geometry_Optimization_1`.

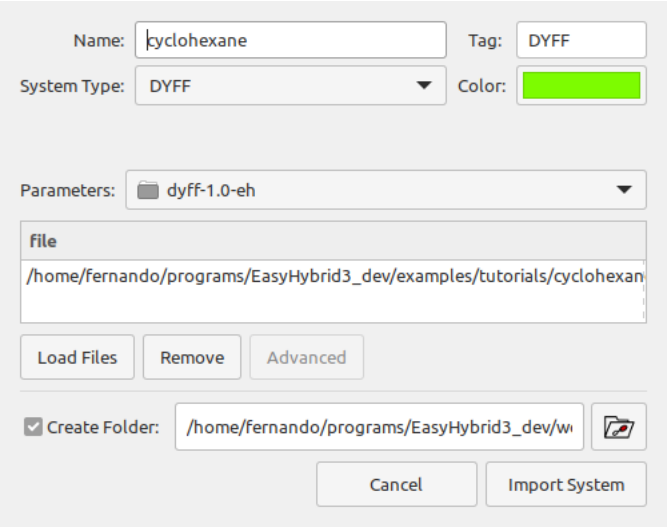


Figure 1. Cyclohexane with the DYFF force field.

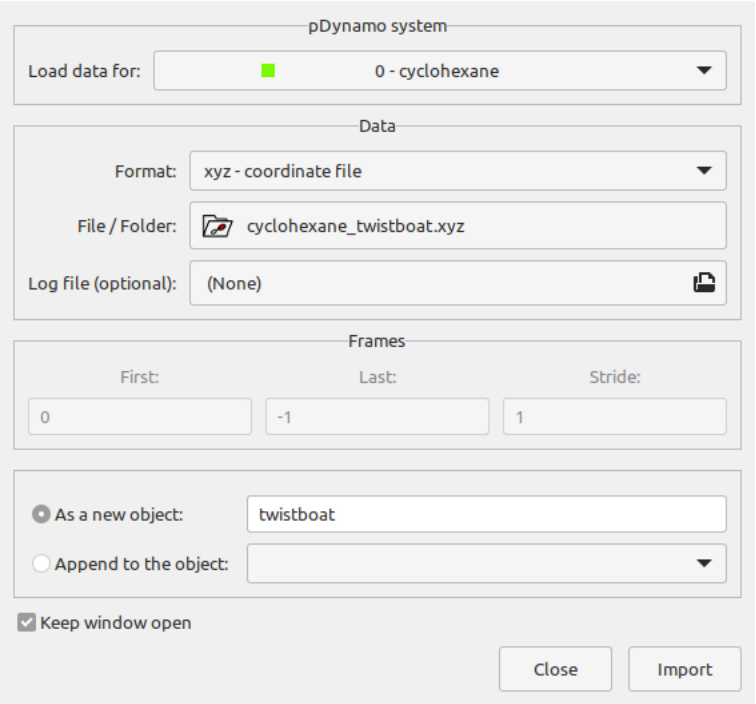


Figure 2. Importing other coordinates of the same molecule.

Both end points must be minima of the same energy function: optimize them before the NEB.

2. Run the NEB

Simulate → Nudged Elastic Band (NEB) (Figure 3):

Field	Value	Meaning
Reactants	`0_Geometry_Optimization_0` (chair)	first image
Products	`0_Geometry_Optimization_1` (twist-boat)	last image
Fixed Terminal Images	on	the end images are not optimized
Number of Structures	11	9 images between the ends
Maximum Iterations / RMS Grad. Tolerance	1000 / 0.1	convergence of the chain
Spring Force Constant	500	keeps the images apart along the path
Use Spline Redistribution	on (default)	keeps the images evenly spaced
Trajectory Name	`cyclohexane_NEB`	

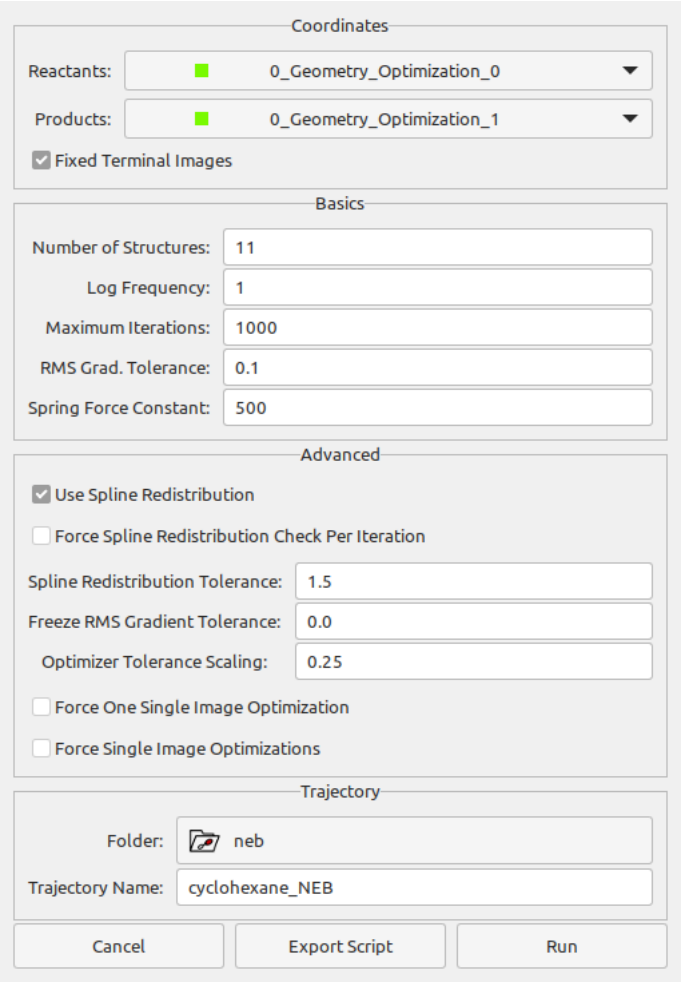


Figure 3. The NEB window.

Click **Run**. The initial chain is built by interpolation between the two ends and then optimized; here it converges in 46 iterations (a few seconds). Import the result from the *Finished!* dialog: an object with 11 frames.

3. The path

Analysis → PES Analysis with the NEB object (Figure 4). The x axis is the image (or the distance along the path, **Profile x**).

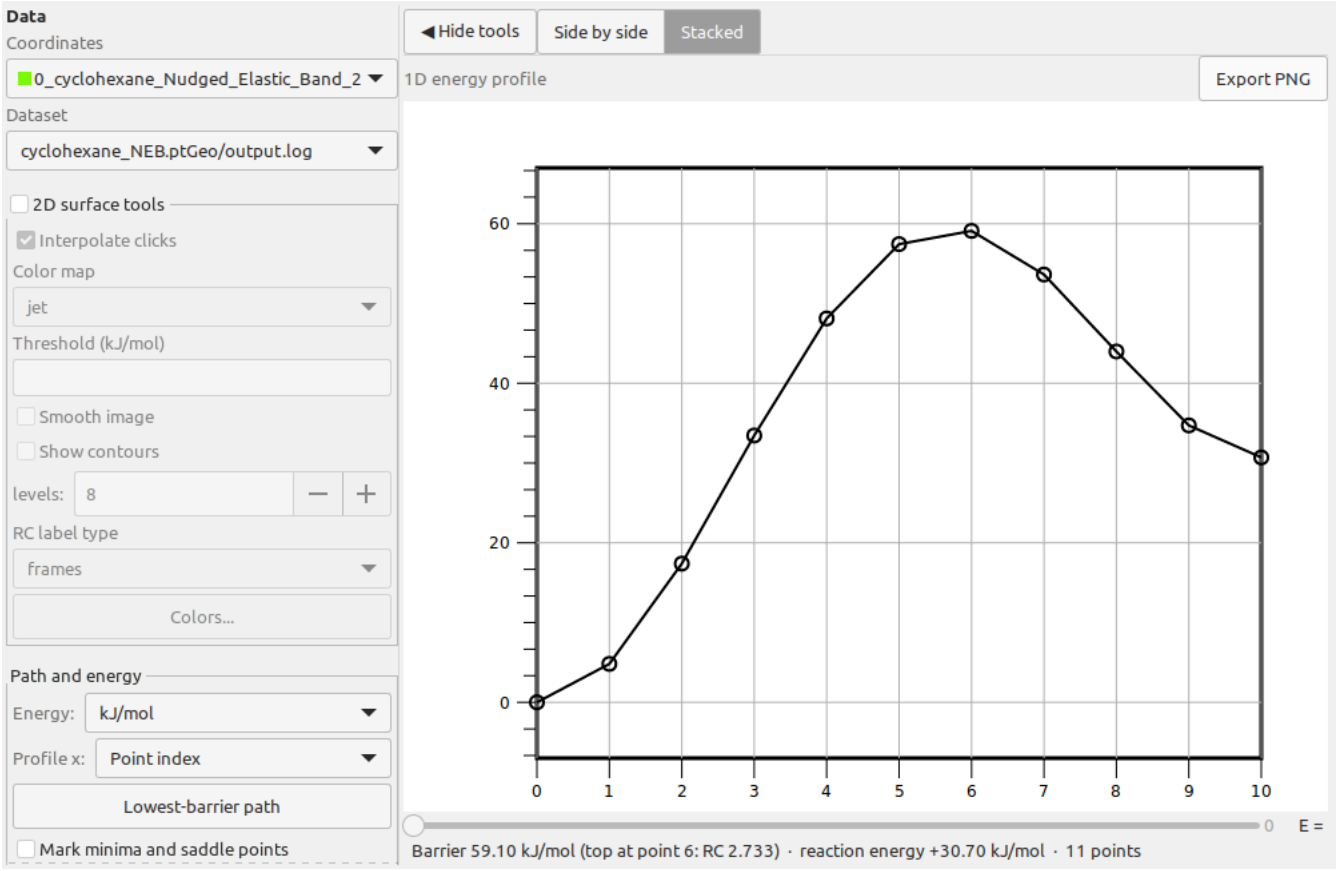


Figure 4. Energy along the chair → twist-boat path.

	Image	E (kJ/mol)	Experiment
Chair	0	0.0	
Highest image (half-chair region)	6	**59.1**	about 45 (10.8 kcal/mol)
Twist-boat	10	30.7	about 23 (5.5 kcal/mol)

The ring torsions show the change: ±57° alternating in the chair, (30°, 30°, −63°) twice in the twist-boat. The highest image has three consecutive torsions close to zero (−6°, −16°, −6°): one side of the ring is flattened, the half-chair shape of the transition state of the ring flip (Figure 5).

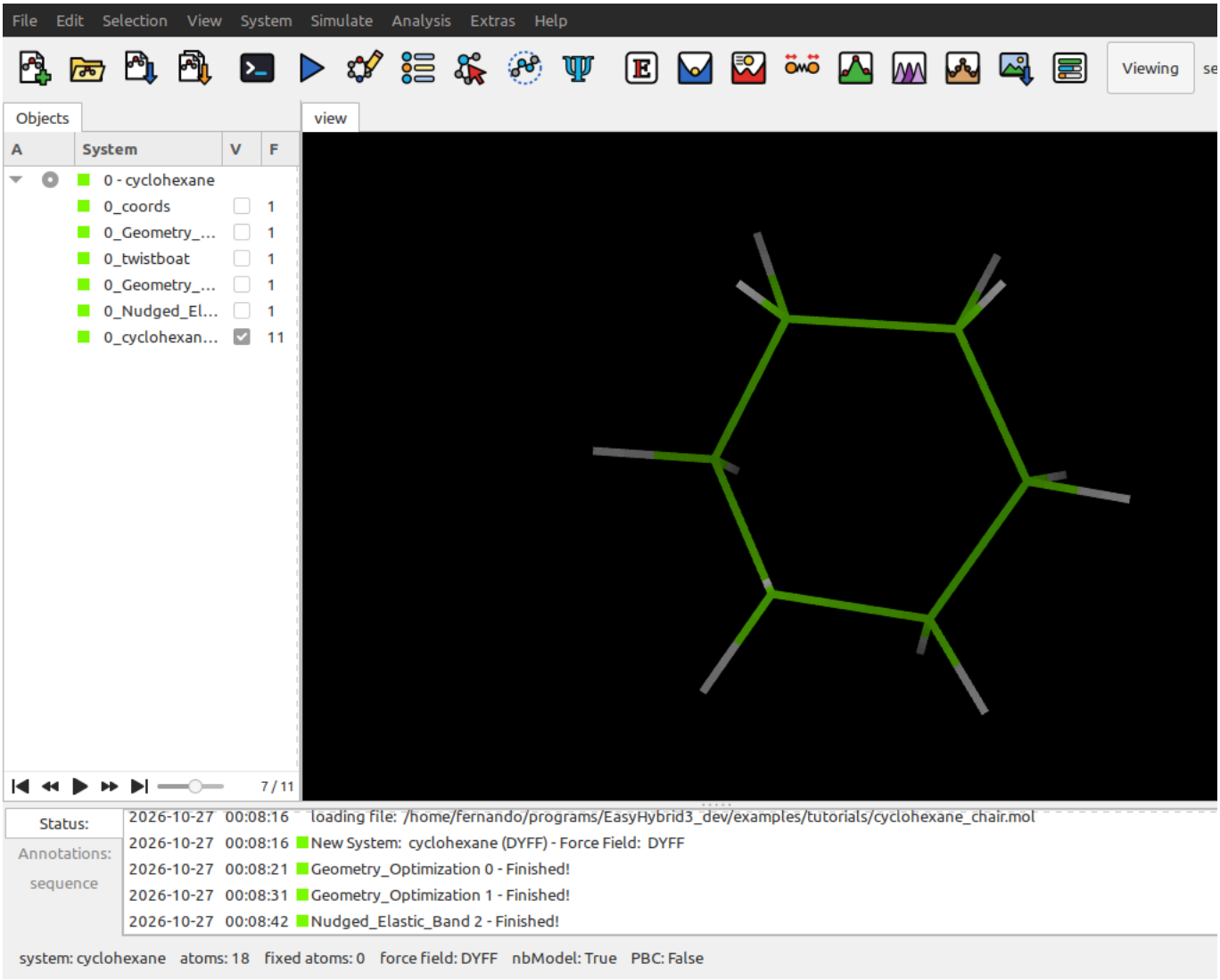


Figure 5. Image 6, the highest point of the path.

The force field matters. DYFF overestimates the barrier by about 14 kJ/mol here. With the small OPLS set that pDynamo provides for its book examples (`bookSmallExamples`), the same calculation gives a twist-boat 62 kJ/mol and a barrier 97 kJ/mol above the chair: that set is meant for teaching the methods, not for conformational energies.

Notes

- **More images** describe the top better; with few images the highest image may be far from the true saddle point. Refine it with **Simulate → Search Transition State (Baker)** and check it with **Simulate → Normal Modes** (one imaginary frequency).
- **The end points must differ only where the reaction happens.** In an enzyme, reactant and product optimized separately can differ in the arrangement of many water molecules and side chains; the NEB then follows those changes too and may not converge. Taking the ends from a scan, or from structures relaxed from each other, avoids it (see Chorismate mutase (2): NEB).
- References: G. Mills, H. Jónsson, *Phys. Rev. Lett.* **1994**, 72, 1124 (the NEB method); F. A. L. Anet, A. J. R. Bourn, *J. Am. Chem. Soc.* **1967**, 89, 760 (barrier of the cyclohexane ring inversion).

EasyHybrid 3.2.0 · built 2026-10-29 from `dev` @ `3539da4` · https://github.com/ferbachega/EasyHybrid3